Research Quality Guidelines for AI and Data Science

Academic Research

- The research report is designed to support the development and evaluation of the primary deliverables (e.g., algorithms, systems, models).
- Research must provide evidence-based insights or evaluations that directly support decision-making or validation of the main deliverables.

Research Goals

- Clearly defined objectives that align with the challenges and opportunities within AI and Data Science fields.
- Goals must integrate considerations for practical implementation and/or theoretical advancement.

Background or Literature

- Sources must include a mix of academic research and industry white papers or case studies.
- Context must be provided regarding current AI or Data Science challenges, theoretical frameworks, and methodologies.
- A clear link between the literature and the research goals should be established.

Research Methods

- Methods must be aligned with the research objectives and tailored to the specific challenges of Data Science and AI. These can include data collection techniques , exploratory data analysis, feature engineering, machine learning model development, deep learning methods, natural language processing techniques, and evaluation metrics. Students may also incorporate qualitative methods such as interviews, focus groups, and content analysis to complement quantitative approaches, providing context or grounding for AI models. A clear rationale for the choice of methods must be provided, along with evidence of efforts to uncover insights that go beyond readily available or surface-level information.
- Methods should be justified with respect to the research questions and goals, incorporating techniques such as machine learning, data visualization, statistical modeling, or causal inference.

Findings (Results) and Discussion

- Findings must be structured and presented systematically
- Results should address the research questions, linking back to the theoretical concepts and challenges identified in the literature.
- Discussions should include limitations of the methods and datasets used and identify opportunities for future work.

Conclusion

- Conclusions must offer actionable recommendations, such as system improvements, design changes, or deployment strategies.
- Recommendations should address both industry needs and academic contributions, ensuring relevance to AI and Data Science communities.

Language and Layout

- Reports must be written in clear, formal, academic English.
- Layout should include proper use of headings, tables, and a detailed table of contents.
- References must adhere to APA (7th edition) citation standards.

Sections for Graduation Project Reports

- **Abstract:** A concise summary of the research goals, methods, findings, and conclusions.
- **Acknowledgments:** A section to thank supervisors, contributors, and institutions that supported the research.
- **Introduction:** A detailed introduction providing an overview of the problem, research objectives, and significance of the study.
- **Problem Statement:** A clear articulation of the problem being addressed, supported by evidence or industry relevance.
- **Methodology:** Comprehensive details about the research design, data collection methods, tools, and frameworks used.
- **Results:** A dedicated section presenting data, charts, and other visuals to illustrate findings.
- **Discussion:** An analysis of the results, comparisons with prior research, and implications of the findings.
- **Future Work:** Suggestions for extending the research or applying it to other domains.
- **Appendices:** Supporting documents, raw data, or additional details that do not fit into the main sections.

By adhering to these guidelines, researchers in AI and Data Science can ensure their work meets high-quality standards, is impactful, and contributes meaningfully to the field.